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SUPERSEDING
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(See section 6)

MILITARY SPECIFICATION

SHELL BLANKS; STEEL, HIGH-TENSILE (TORPEDOES)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers high-tensile shell blanks fabricated from sheet steel for use in torpedo components.

1.2 Classification. Shell blanks shall be of the following types as specified (see 6.2):

Type I - Pieced and welded
Type II - Seamless and drawn or spun.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents of the issues in effect on the date of invitation for bids or request for proposals form a part of this specification to the extent specified herein. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

SPECIFICATION

Military

MIL-C-16173	Corrosion Preventive Compound, Solvent Cutback, Cold-Application

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STANDARDS

Federal

FED-STD-151	Metals, Test Methods

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

*2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply. In the event of conflict between this specification and other documents referenced herein, requirements of this specification shall apply.

American Society of Mechanical Engineers (ASME)

ASME Boiler and Pressure Vessel Code - Section VIII, Rules For Construction of Pressure Vessels.

(Copies of ASME publications may be obtained from the American Society of Mechanical Engineers, 345 East 47th Street, New York City, N.Y. 10017.)

3. REQUIREMENTS

*3.1 Material. Shell blanks, type I and type II, shall be manufactured of deep-drawing weldable quality, high strength, low alloy, hot-rolled sheet steel.

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3.1.1 Chemical composition. Steel used in high-tensile shell blanks shall conform to the following chemical analysis:

Carbon	0.15% maximum
Manganese	1.00% maximum
Sulfur	0.05% maximum
Alloying elements	As approved

3.1.1.1 Bidders shall state the chemical composition they propose to furnish. Such chemical data shall include the maximum and minimum content of all pertinent elements present. The right is reserved to reject any bid on a composition which is considered unfavorable as to properties or availability of alloys.

3.2 Construction.

3.2.1 Type I shell blanks. Type I shell blanks shall be pieced and welded. Welds shall be of the open square butt type. The weld metal shall be clean and sound. Fusion of the base and the deposited metal shall be complete at all points in the weld. Welded bead metal shall completely fill the required space and shall be removed smooth and flush with adjacent surfaces, both inside and outside the shell. Welds shall be free from discontinuities, cracks, porosity, and other defects which would affect the serviceability of the shell blanks when subjected to pneumatic pressure. Neither welding rod nor flux used shall be such as to cause objectionable corrosive action with the shell material.

3.2.2 Type II shell blanks. Type II shell blanks shall be seamless drawn or spun.

3.3 Mechanical properties.

3.3.1 Formed steel. Formed steel in finished shell blanks shall conform to the following mechanical properties:

Tensile strength	70,000 psi (minimum)
Yield point (0.5% stretch)	50,000 psi (minimum)
Elongation in 2 inches	22% (minimum)
Bend test	Shall not crack when tested as specified in 4.6.3.

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3.3.2 Weld joints. Weld joints in finished type I shell blanks shall conform to the following mechanical properties:

Tensile strength	70,000 psi (minimum)
Bend test	Shall not crack when tested as specified in 4.6.3.

3.4 Metallographic structure. The finished shell shall be uniform and free from defects and excessive impurities.

3.5 Dimensions. Dimensions and tolerances shall conform to the drawings specified in the contract or order.

3.6 Finish. Shell blanks shall be clean, smooth, and free from dents, bulges, pits, cracks, scale, and other injurious defects.[1]

3.7 Marking. Each shell blank shall be plainly and permanently marked with the steel mill heat number, contract number, manufacturer's trademark, and inspector's symbol. This information shall be stamped on the inside of the shell blank 4.5 inches ± 0.5 inch from the open end. Marking shall be not deeper than 0.010 inch.

3.8 Workmanship. Workmanship shall be first class in every respect. Shell blanks which have been repaired by welding or soldering in the body or dome portions shall be rejected.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsibile for the performance of all inspection requirements, as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspection set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and prodecures used in the determination of the acceptability of products shall be in accordance with the provisions of MIL-STD-105.

[1] A uniform condition, normal to a first-class pickle or fine-blast cleaned job, is desired.

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*4.2.1 Samples of first production. Unless otherwise specified, for each contract, a representative sample of two shell blanks shall be selected from the first production lot and forwarded to a laboratory designated by the procuring activity to determine compliance with the requirements of the contract, specification, and drawings.

4.2.2 Sampling for destructive testing.

*4.2.2.1 A sample of one shell blank representative of the lowest quality of the lot (for type I blanks, lowest quality shall be the blanks with the least perfect welds) shall be selected from each lot to determine compliance with the requirements specified herein for destructive testing. The lot shall be rejected for failure of the sample to meet the test requirements.

*4.2.3 Furnishing of samples. Samples required by 4.2.1 and 4.2.2 are to be furnished in addition to the quantity specified in the contract at no additional cost to the Government. Notification of probable date of selection of samples should be made as early as practicable to the procuring activity to allow sufficient time for forwarding of shipping instructions.

4.3 Lot. A lot shall consist of 100 shell blanks manufactured from the same mill heat and under essentially the same conditions.

4.4 Inspection.

*4.4.1 Visual and dimensional inspection. Each of the sample shell blanks selected in accordance with 4.2 shall be visually and dimensionally inspected to determine compliance with the requirements of this specification. Any shell blank in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective shell blanks in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

*4.4.2 Rejected lots. Rejected lots may be offered again for inspection and test, provided that all nonconforming shell blanks have been repaired or removed. Samples shall be again, selected and examined from such resubmitted lots to determine compliance with this specification.

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4.5 Classification of tests.

4.5.1 First production. The tests specified under 4.5.2 shall be performed on sample shell blanks selected in accordance with 4.2.1.

4.5.2 Production tests. The following tests shall be performed on samples:

a. Chemical analysis, 4.6.1
b. Mechanical properties, 4.6.2
c. Bend test, 4.6.3
d. Metallographic structure, 4.6.4
e. Radiographic test, 4.6.5.

4.6 Tests.

4.6.1 Chemical analysis. Chips for chemical analysis shall be taken from the sample shell blank selected to represent the lot for test purposes. Chips shall be taken from sound metal. The sample shall consist of not less than 2 ounces of chips and shall be fine, clean, and free from oil, dirt, grit, or other foreign matter. The chips shall be forwarded to a Government laboratory for analysis to determine compliance with composition requirements of 3.1.1.

*4.6.2 Mechanical properties (except bend test). Specimens shall be taken from the sample shell blank and may be flatened if required. They shall be prepared and tested in accordance with FED-STD-151 to determine compliance with 3.3.

4.6.3 Bend test (cold).

4.6.3.1 Two test specimens, 1/2 inch wide by 4 inches long with cut edges ground and corners broken on approximately 1/32-inch radius, shall be taken from the sample shell blank to represent each piece of sheet steel and each welded joint used in the fabrication of the shell. Weld, when present, shall be centrally located and at right angles to the long axis of the specimen.

4.6.3.2 The diameter of the mandrel used in making this test shall be twice he thickness of the specimen.

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4.6.3.3 Of the two specimens for any single sheet or weld, one shall be bent forward 180° over the mandrel to place the original outside surface in tension and the other bent in reverse 180° over the mandrel to place the original inside surface in tension. Cold bend test for type I and type II shells shall meet the requirements of 3.3.

*4.6.4 Metallographic structure. The test for metallographic structure shall be in accordance with FED-STD-151 to determine compliance with 3.4.

4.6.4.1 Macroetch. Specimens not less than 0.5 inch wide by 4 inches long shall be taken from the sample shell. Cut edges of specimens shall be ground and polished. Specimens etched for 10 minutes, in 1:1 hydrochloric acid and water, at 180° F shall be found by visual inspection to be free from gas holes, laminations, seams, or other defects.

4.6.4.2 If macroetch reveals conditions difficult to evaluate, samples shall be polished etched with 5 percent Nital and examined at 100 magnifications. Excessive segregation or abnormalities shall be cause for rejection.

*4.6.5 Radiographic test. Samples of type I shell blanks shall be subjected to radiographic inspection of welds in accordnance with ASME Boiler and Pressure Vessel Code, Section VIII.

*4.6.6 Dimensions. The shell blanks shall be checked to determine compliance with the dimensions and tolerances specified on the drawing.

4.7 Inspection. Each shell blank shall be inspected to determine compliance with the requirements for construction, marking and workmanship. Packaging shall be examined for compliance with the requirements of section 5.

4.8 Destruction of material. All samples destroyed in making tests required by this specification shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increasing the cost of same.

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5. PREPARATION FOR DELIVERY

5.1 Preservation. Unless otherwise specified, steel shell blank surfaces shall be coated with corrosion-preventive compound conforming to MIL-C-16173, grade 2.

5.2 Packing. Unless otherwise specified, shell blanks shall be boxed, crated, placed on skids, or otherwise protected from mechanical damage, as the size may require, to insure acceptance for safe delivery by common or other carrier to the point of delivery at the lowest rate.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. High-tensile steel shell blanks covered by this specification are intended for use as torpedo components.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification
b. Type of shell blank required
c. Applicable drawings
d. Special markings.

6.3 Supersession data. This specification includes the requirements of NAVORD OS 2519A dated 10 November 1943.

*6.4 The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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